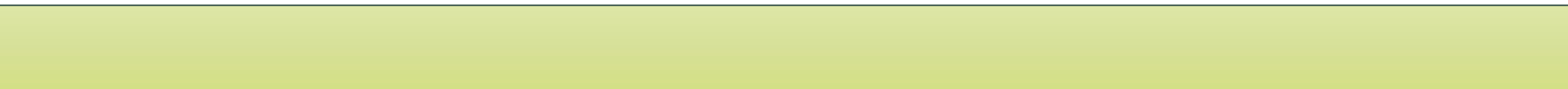


Image Derivatives & Edge Detection



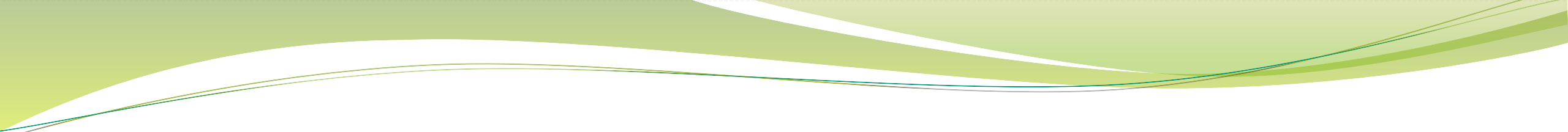
Sharpening Spatial Filters

Previously we have looked at smoothing filters which **remove fine detail**

Sharpening spatial filters seek to **highlight fine detail**

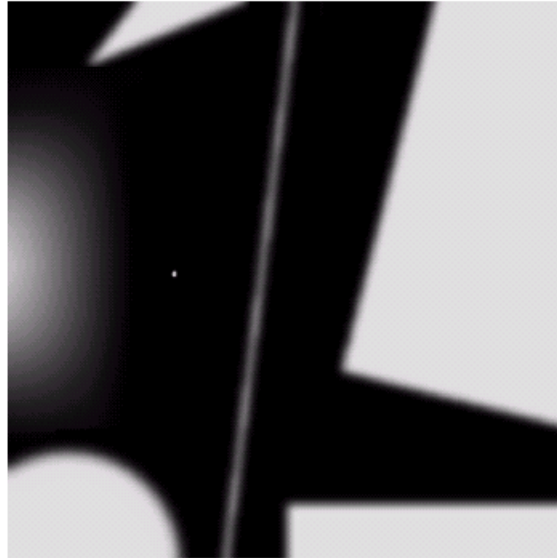
- Remove blurring from images
- Highlight edges

Sharpening filters are based on ***spatial differentiation***

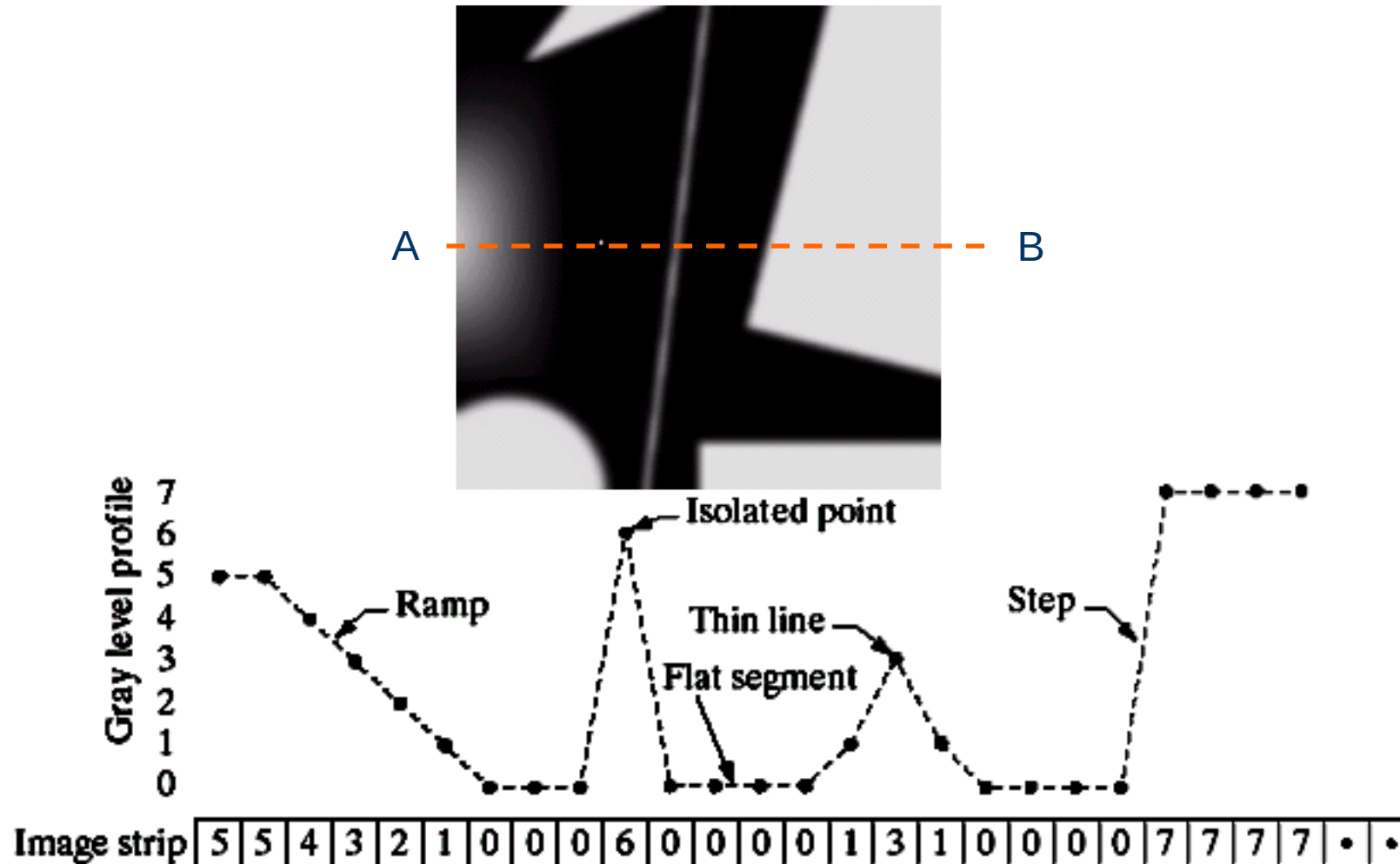
- 
- Sharpening filters are based on first- and second-order derivatives of image.
 - The derivatives of a digital function are defined in terms of differences.

Spatial Differentiation

- Let's consider a simple 1 dimensional example



Spatial Differentiation



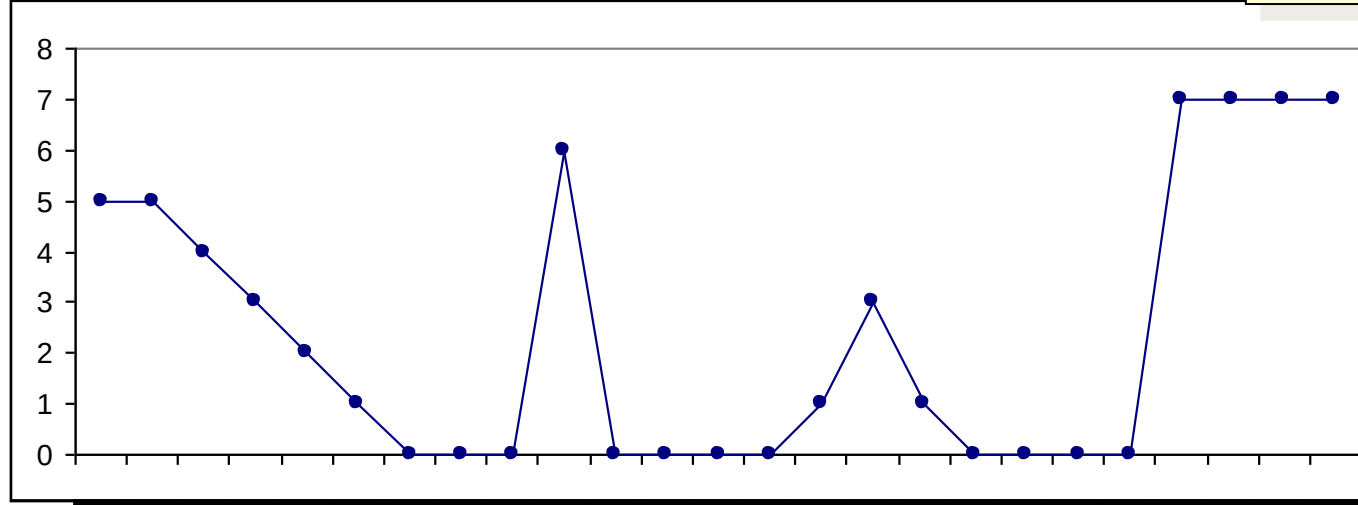
1st Derivative

- The 1st derivative of a function is given by:

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

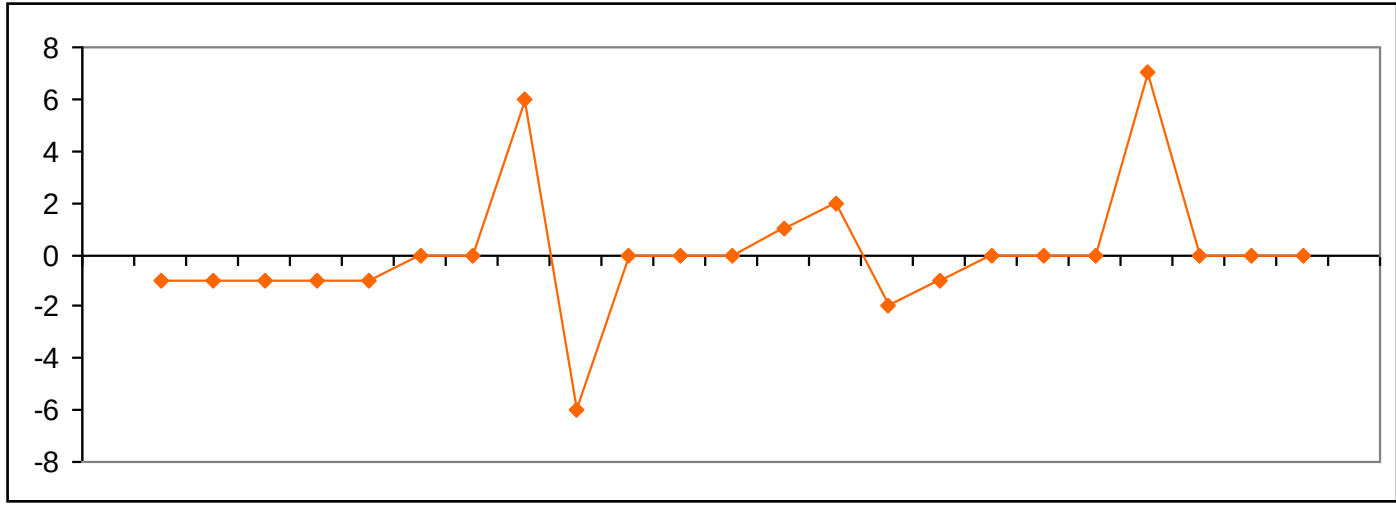
- Its just the difference between subsequent values and measures the rate of change of the function

1st Derivative



5	5	4	3	2	1	0	0	0	6	0	0	0	0	1	3	1	0	0	0	0	7	7	7	7
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

	-1	-1	-1	-1	-1	0	0	6	-6	0	0	0	1	2	-2	-1	0	0	0	7	0	0	0	
--	----	----	----	----	----	---	---	---	----	---	---	---	---	---	----	----	---	---	---	---	---	---	---	--



Definition for a first derivative

- ❑ Must be zero in flat segments
- ❑ Must be nonzero at the onset of a gray-level step or ramp
- ❑ Must be nonzero along ramps
- ❑ A basic definition of the first-order derivative of a one-dimensional function $f(x)$ is

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

(Diff b/w subsequent values & measures the rate of change of the function)

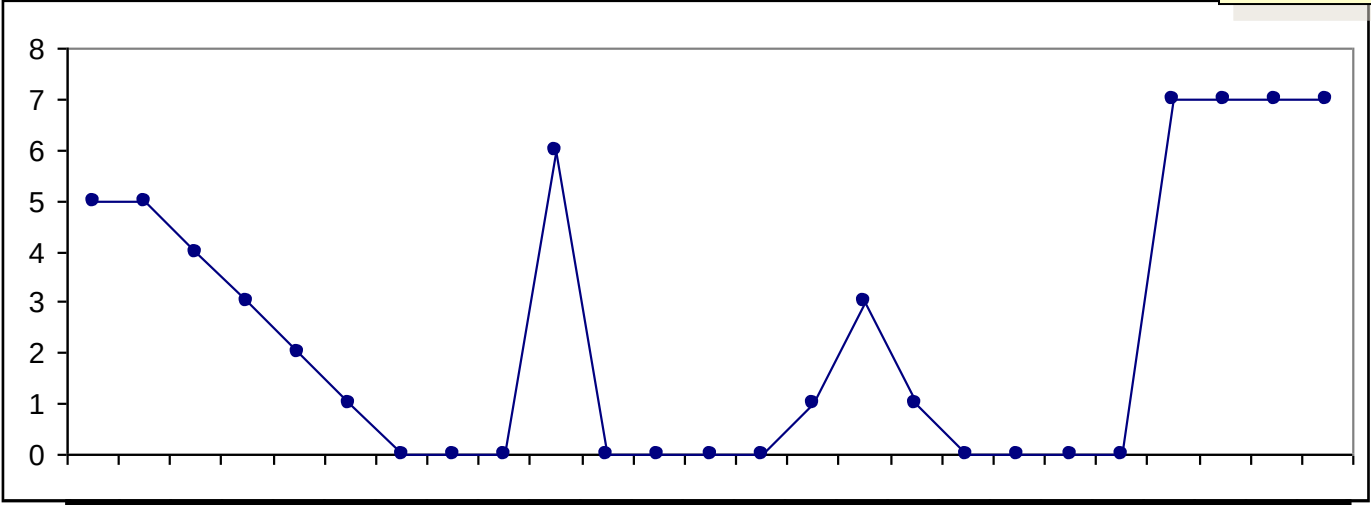
2nd Derivative

The 2nd derivative of a function is given by:

$$\frac{\partial^2 f}{\partial^2 x} = f(x+1) + f(x-1) - 2f(x)$$

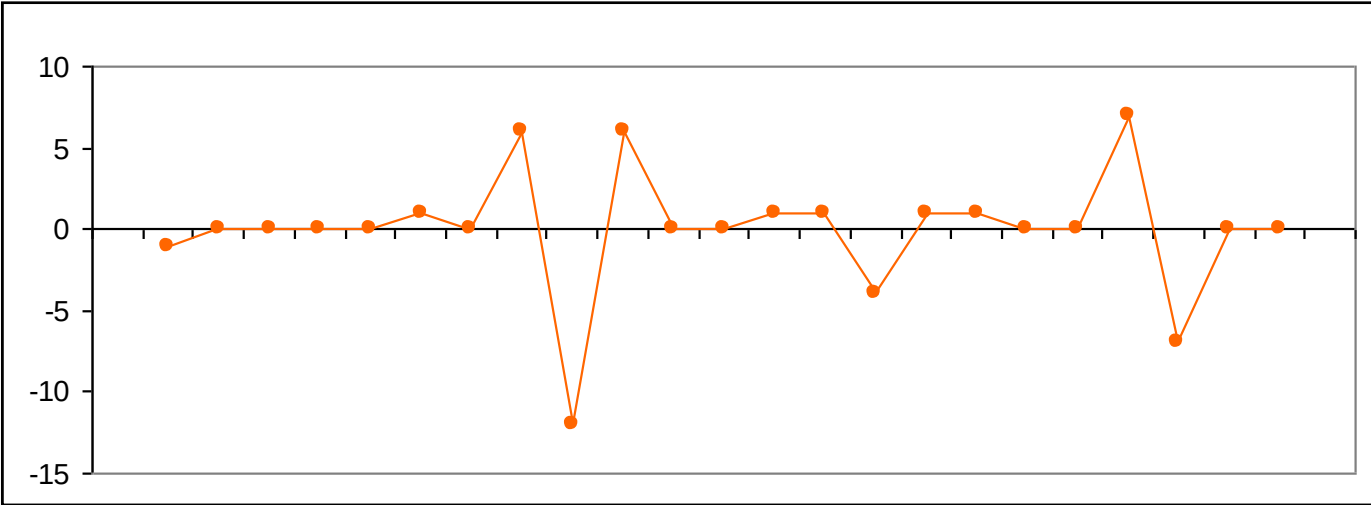
Simply takes into account the values both before and after the current value

2nd Derivative



5	5	4	3	2	1	0	0	0	6	0	0	0	0	1	3	1	0	0	0	0	7	7	7	7
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

	-1	0	0	0	0	1	0	6	-12	6	0	0	1	1	-4	1	1	0	0	7	-7	0	0	
--	----	---	---	---	---	---	---	---	-----	---	---	---	---	---	----	---	---	---	---	---	----	---	---	--



Definition for a second derivative

- ❑ Must be zero in flat areas
- ❑ Must be non zero at the onset and end of a gray-level step or ramp
- ❑ Must be zero along ramps of constant slope
- ❑ We define a second-order derivative as the difference

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$

(the values both before & after the current value)

2nd Derivative for Image Enhancement

The 2nd derivative is more useful for image enhancement than the 1st derivative - *Stronger response to fine detail*

We will come back to the 1st order derivative later on

The first sharpening filter we will look at is the ***Laplacian***

Laplacian Filter

The Laplacian is defined as follows:

$$\nabla^2 f = \frac{\partial^2 f}{\partial^2 x} + \frac{\partial^2 f}{\partial^2 y}$$

$$\frac{\partial^2 f}{\partial^2 x} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial^2 y} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

Laplacian Filter

So, the Laplacian can be given as follows:

$$\begin{aligned}\nabla^2 f = & [f(x+1, y) + f(x-1, y) \\ & + f(x, y+1) + f(x, y-1)] \\ & - 4f(x, y)\end{aligned}$$

Can we implement
it using a filter/
mask?

0	1	0
1	-4	1
0	1	0

Laplacian Filter

0	1	0	1	1	1
1	-4	1	1	-8	1
0	1	0	1	1	1

0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

a	b
c	d

FIGURE 3.39

(a) Filter mask used to implement the digital Laplacian, as defined in Eq. (3.7-4).

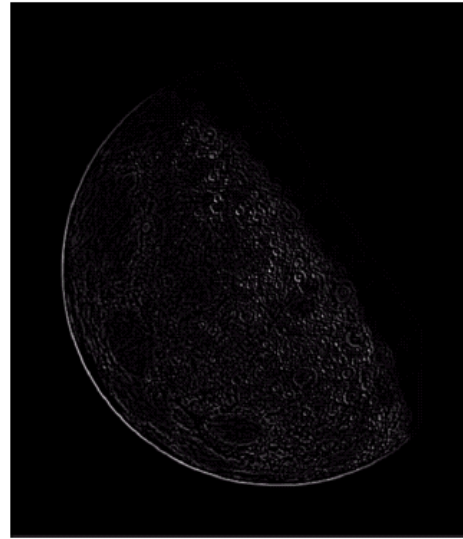
(b) Mask used to implement an extension of this equation that includes the diagonal neighbors. (c) and (d) Two other implementations of the Laplacian.

Laplacian Filter

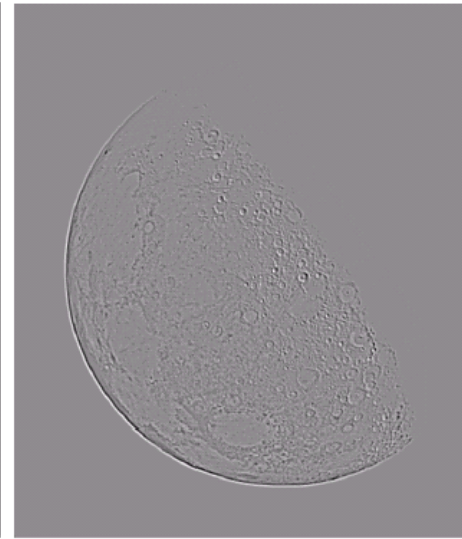
Applying the Laplacian to an image we get a new image that highlights edges and other discontinuities



Original
Image



Laplacian
Filtered Image

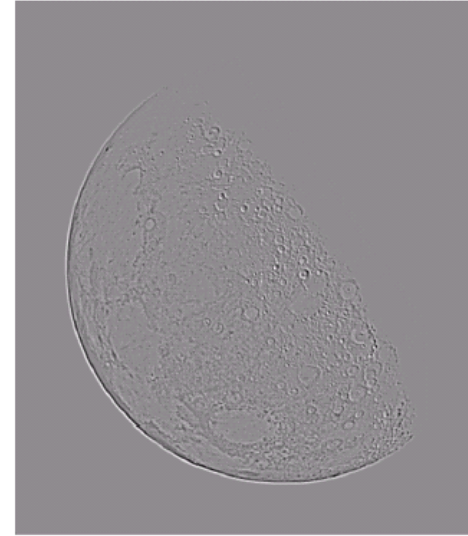


Laplacian
Filtered Image
Scaled for Display

Laplacian Image Enhancement

The result of a Laplacian filtering is not an enhanced image

To generate the final enhanced image

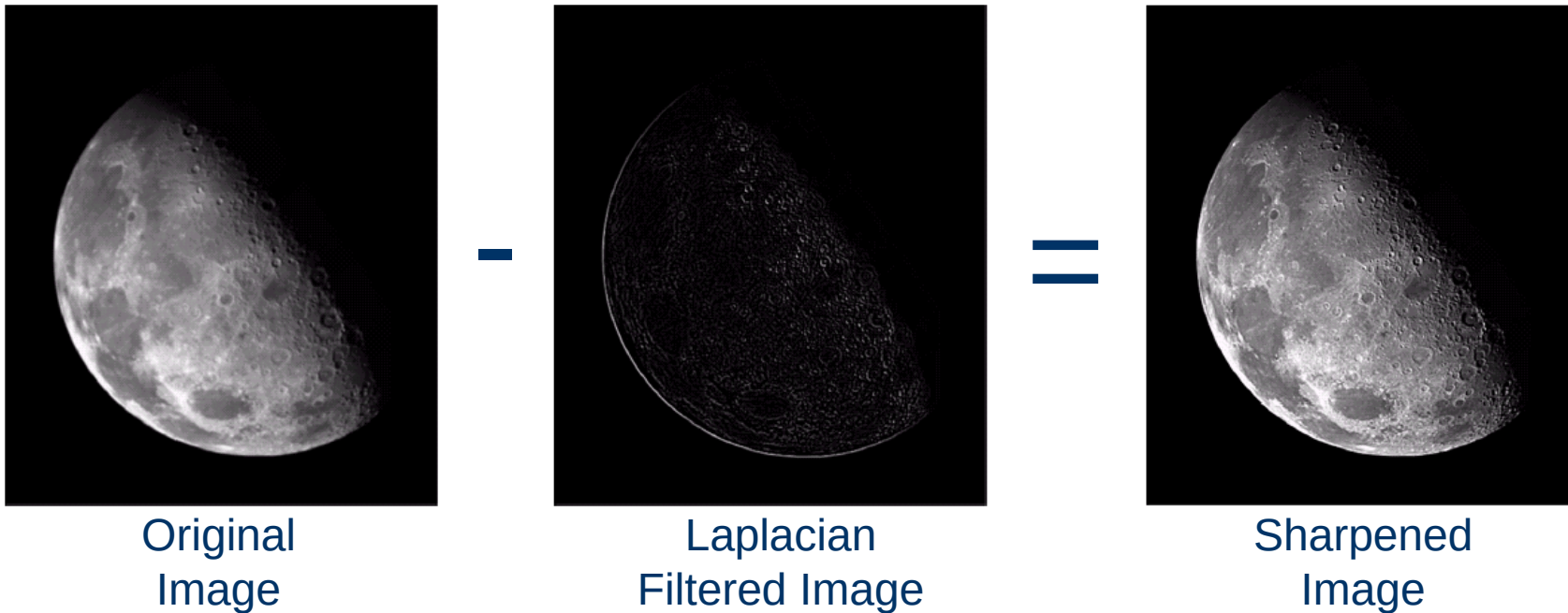


Laplacian
Filtered Image
Scaled for Display

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f, & w_5 < 0 \\ f(x, y) + \nabla^2 f, & w_5 > 0 \end{cases}$$

Laplacian Image Enhancement



In the final sharpened image edges and fine detail are much more obvious

Laplacian Image Enhancement

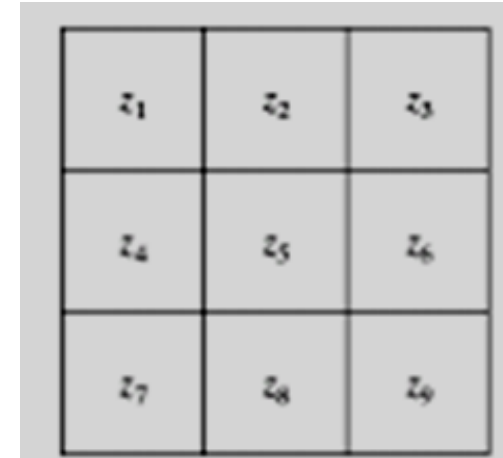


Gradient Operators

Prewitt Operator

□ is used for detecting edges horizontally and vertically.

$$\nabla f \approx \left| (z_7 + z_8 + z_9) - (z_1 + z_2 + z_3) \right| \\ + \left| (z_3 + z_6 + z_9) - (z_1 + z_4 + z_7) \right|$$



z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

Pixel Arrangement

$$\frac{\partial f}{\partial y} = \begin{array}{|c|c|c|} \hline -1 & -1 & -1 \\ \hline 0 & 0 & 0 \\ \hline 1 & 1 & 1 \\ \hline \end{array}$$

Extract horizontal edges

$$\frac{\partial f}{\partial x} = \begin{array}{|c|c|c|} \hline -1 & 0 & 1 \\ \hline -1 & 0 & 1 \\ \hline -1 & 0 & 1 \\ \hline \end{array}$$

Extract vertical edges

Gradient Operators

Sobel Operator

$$\frac{\partial f}{\partial y} =$$

-1	-2	-1
0	0	0
1	2	1

Extract horizontal edges

$$\frac{\partial f}{\partial x} =$$

-1	0	1
-2	0	2
-1	0	1

Extract vertical edges

Emphasize more the current point
(x direction)

$$\nabla f \approx |(z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)| \\ + |(z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)|$$

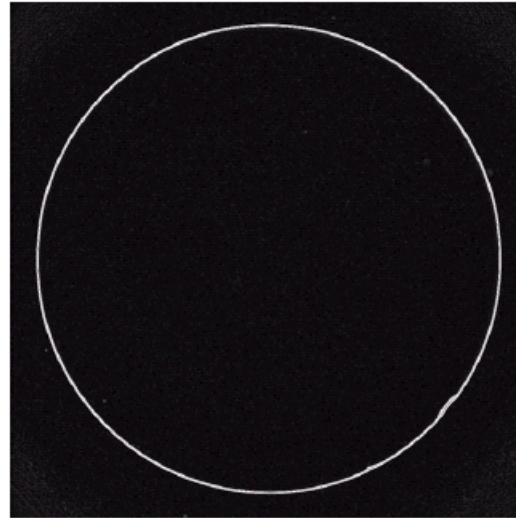
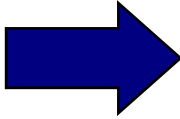
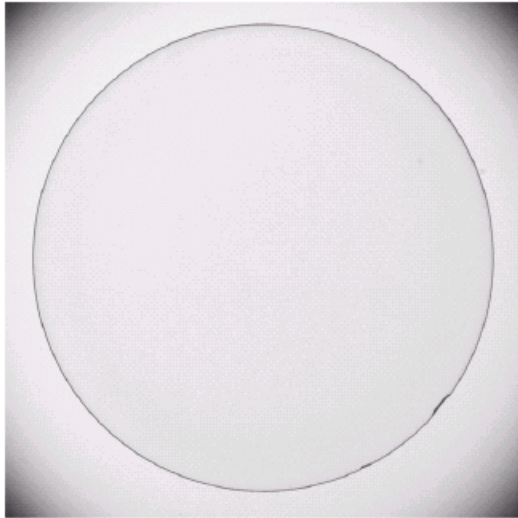
Emphasize more the current point (y
direction)

Same as Prewitt Operator

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

Pixel Arrangement

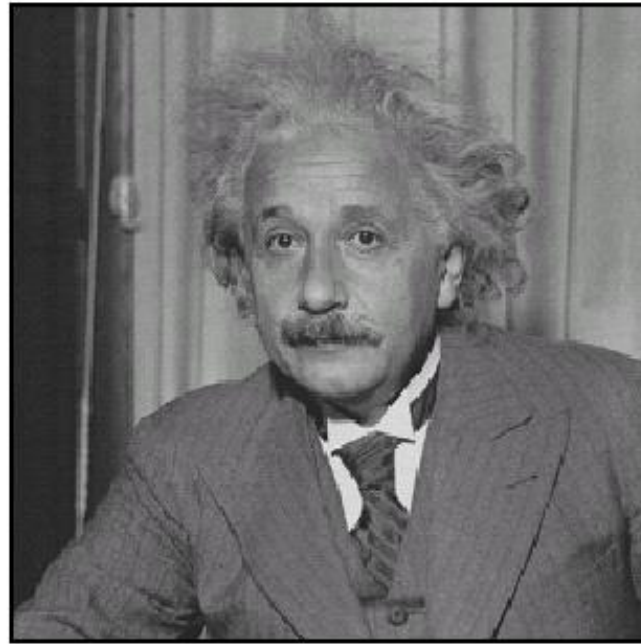
Sobel Operator: Example



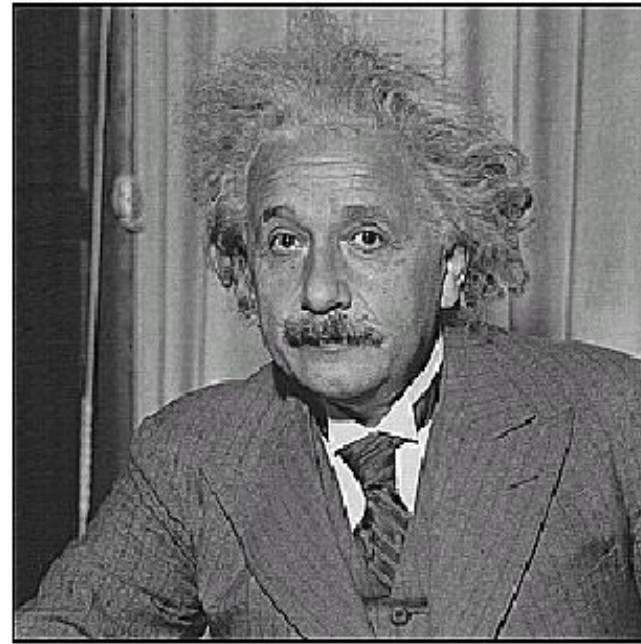
An image of a contact lens which is enhanced in order to make defects more obvious

Sobel filters are typically used for edge detection

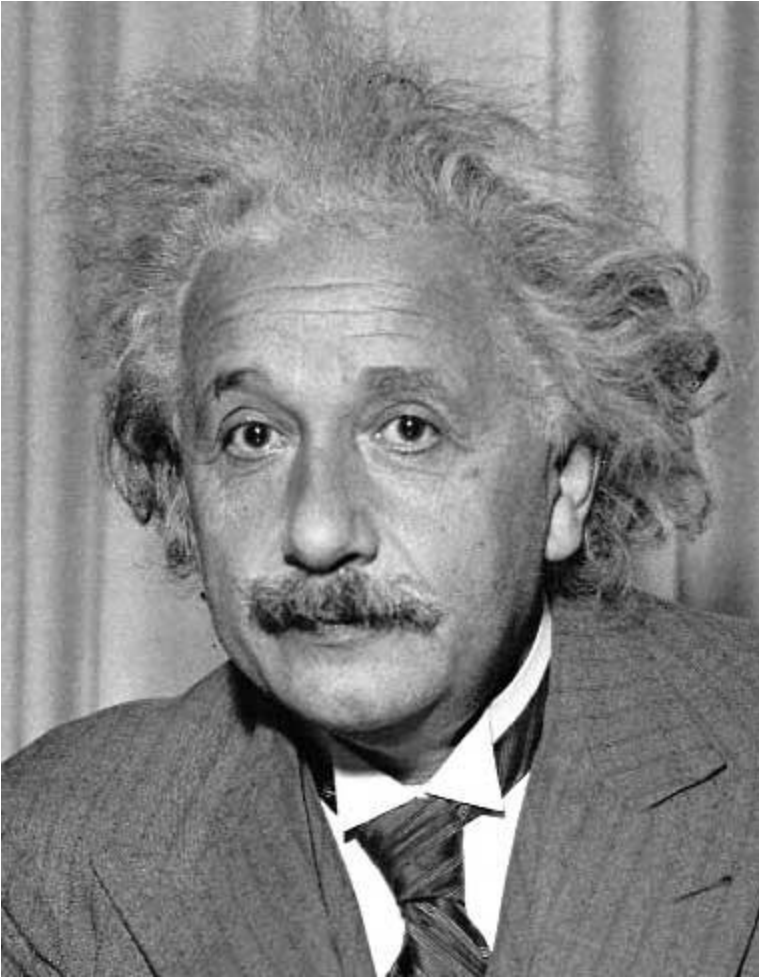
Sharpening



before

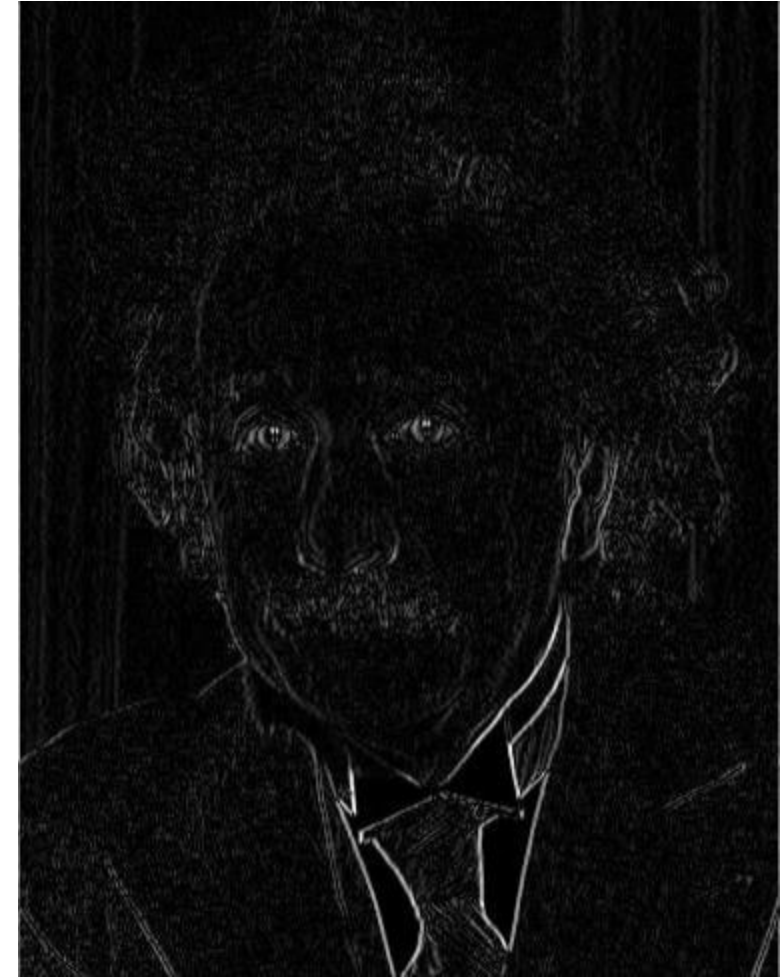


after

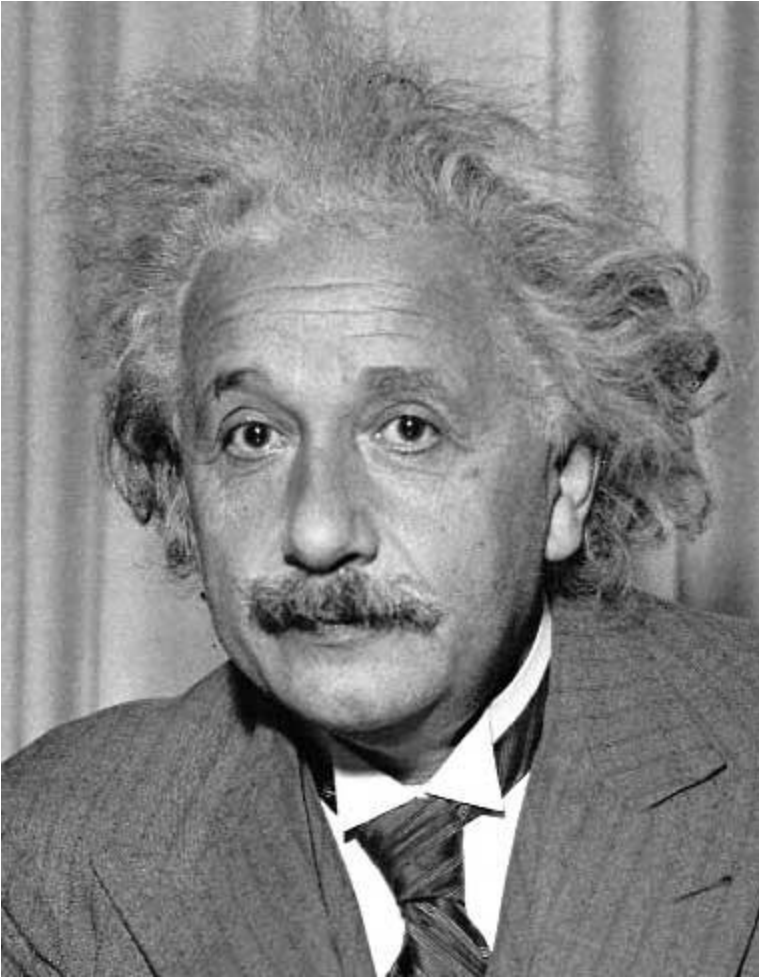


1	0	-1
2	0	-2
1	0	-1

Sobel

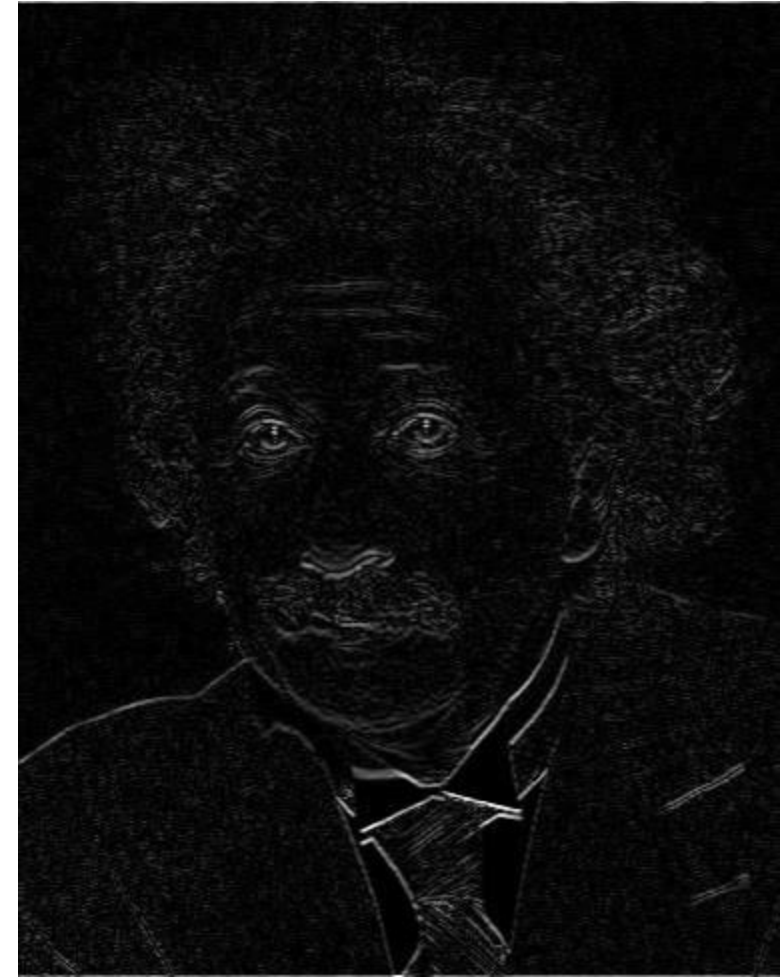


Vertical Edge
(absolute value)



1	2	1
0	0	0
-1	-2	-1

Sobel



Horizontal Edge
(absolute value)